

PRD: Corrective line edits via transcript

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Date created (or last modified): Apr 19, 2024

Status: 100% Launched! ▾

Description:

Provide learners with corrective “line edits” while reviewing their lesson recordings and materials.

Problem / Why / Hypothesis:

What is the user problem we’re trying to solve?

- Students desire more feedback and often express frustration when their tutors don’t correct their English “enough” during or after their lessons
 - [Mauricio](#): “So right here - correctional style after every mistake. I can choose between those options. But ...I don't think they can read it on their end. I don't think that they know about my preferences. ...I would write here to send them. I want to be corrected after every mistake. This part - stop me. I want you to stop me. But I think they don't have that information I can put right here, ...If that information was given to them. Yes.”
 - [Mohammed](#): “...like some tutors, they explain your mistakes in a detail and he advises you to avoid your mistakes. Some tutors told me if you can't explain a word or something or sentence, change your mind and you could explain it in a different way. ...[And what makes a bad tutor?] ...I do have a lot of mistakes and I know my mistakes, but he don't tell me all of these mistakes and he just skipped the slides. He seems like he doesn't care..”
 - [Deepak](#) wants critical feedback to improve his grammar but doesn’t always get it, and he feels that critical feedback is crucial for him to learn grammar. He watches his lesson recordings to learn to improve.
 - [Minnie's](#) experience of a great lesson was one in which the tutor gave her corrective feedback during the lesson, particularly to help her with her R and L sounds.
- Reviewing past lessons can provide learning reinforcement. See [PRD: Post-Lesson Review \(Mega wishlist\)](#)

What is the business problem we’re trying to solve?

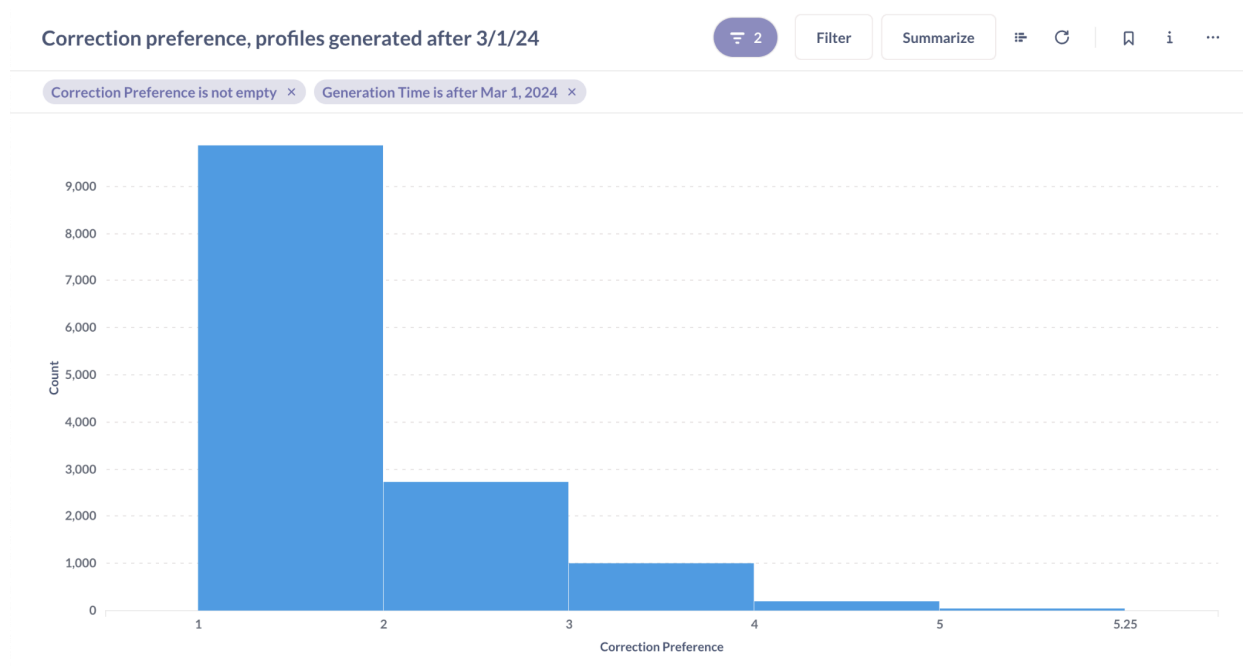
- See [Getting to the Cambly Way](#) , where feedback is an integral part of the learning flywheel that we want to build

What is the problem with the existing situation?

- Learners currently rely on the tutor to provide real-time feedback within the lesson
- Tutors are not always inclined to provide feedback real-time in the lesson because good feedback takes time.
 - [Helen](#): “[Feedback writing is] booked in my breaks at the end of the day, yeah. So it's all extra time, extra work, but sometimes that little bit extra students appreciate

it and they come back. And that's how you get your regulars, just by kind of showing that you care about them, really. It's not just like tick ...\$12, next! I don't kind of work that way."

- [Hal](#): "Cambly doesn't make me waste time writing feedback. Although, guess what? I do write feedback a lot of times and send quick notes, but quick notes to the point that are crucial for that student to hear. ...VIP Kids required something that took an extra 15 minutes. ...you were at a whole different pay scale, if you follow. So I love there's so many good things about Cambly that make it right. One of it is there is no prep time and there is no time after."
- Not every lesson is conducive to providing feedback.
 - For example, it can be challenging to give feedback during Group lessons, [where the tutor may feel uncomfortable giving a single student individual feedback](#). [Learners](#) also understand that the Group lesson is more focused on conversation than on individual corrections.
- [>70% of learners say they want corrections](#) after every mistake during their lesson:
 - 1: "I always want constant corrections in my lesson"
 - 2: "I only want major corrections in my lesson"
 - 3: "I only want corrections at the end of my lesson"
 - 4: "I only want corrections after my lesson"
 - 5: "I don't want corrections at this time"



How will our proposed changes improve the situation?

- Providing the learner with post hoc corrective feedback on things that they could have said differently / correctly (1) addresses the learner's desire for more corrective feedback in a way that doesn't impede the normal flow of conversation and communication ability and (2) helps to cement learning.

Impact:

Who is the user we're solving for?

- Adult Cambly students taking private lessons to start
- (We'll work on feature parity for groups students later)

What % of the population is that segment?

- 100%

How severe is this problem?

- This builds out the post-lesson feedback loop of the learner flywheel
- This also builds out post-lesson review activities (snackactivities!)

How will success appear? Is it a behavioral metric? Is it reduction of a stated user pain point?

- Increased adoption of post-lesson recording / materials review

Changes:

- Identify grammar or vocabulary errors, corrections, and explanations about the error
- Student opts into whether they can see the corrective feedback
- Show the error, correction, and explanation to the learner
- Show grammar and vocabulary corrections that we have high confidence will not show wrong answers
- Prioritize which corrections we show to a learner. This might be based on most frequent errors, or some other heuristic.
- Learner can rate the feedback (negative only)
- Limit to 10 errors

Slack: #corex-transcript-llm-feedback

Figma: [Link to Figma](#)

Tech Spec: Link to tech design doc

Jira: Link to Jira

Dependencies:

N/A

Success Metrics / KPIs:

We will launch at 100%, but with a beta tag

Beta exit criteria:

- <10% of corrections generated rated with a thumbs down

When we will assess Beta:

- 3 weeks post-launch

Instrumentation Needs (PM to assess w/ Eng/PA)

What do we need to measure (including but not limited to Success Metrics)?

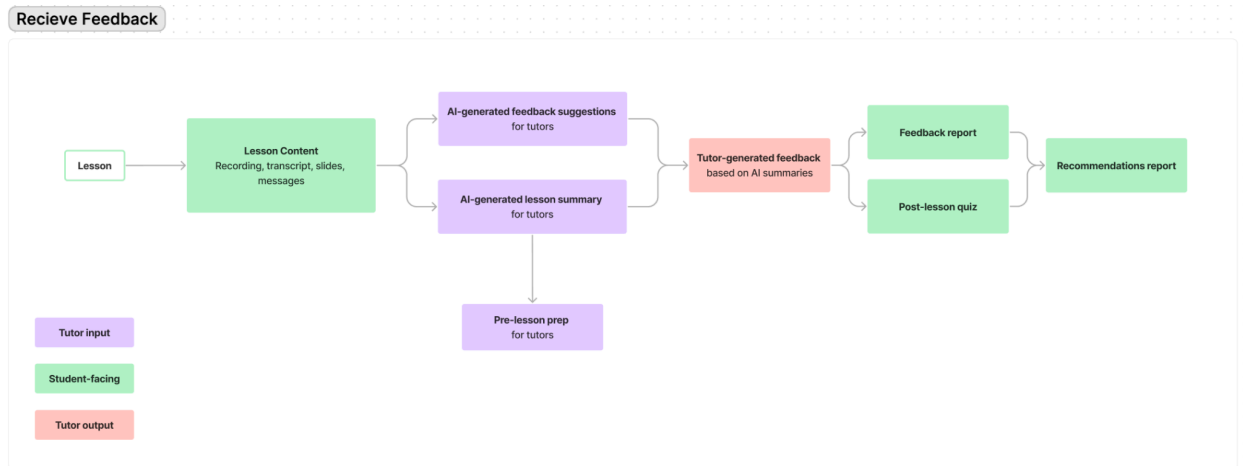
What existing data sources can we derive it from?

What new data sources do we need to create?

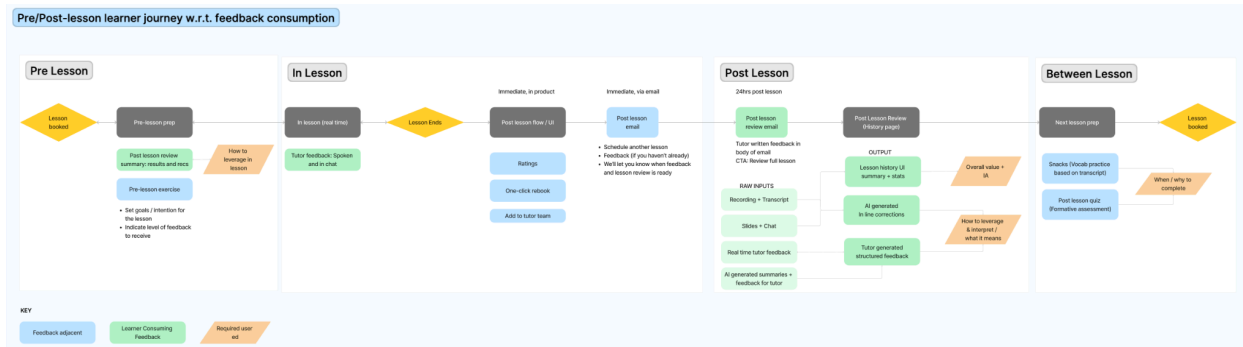
What logging do we need to add?

GTM (PMM to assess):

A cohesive, actionable experience



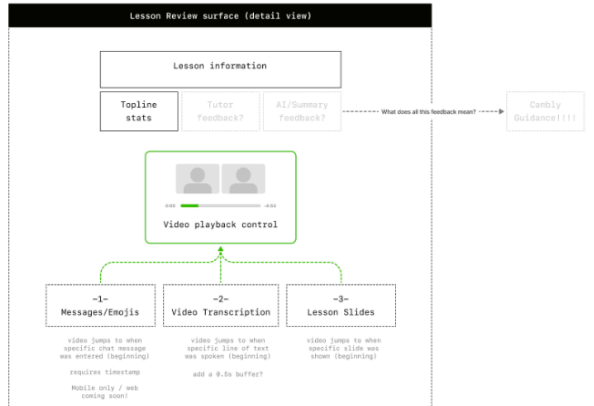
Higher level view (full experience pre-post-pre lesson)



This illustrates a more “self serve” option on the left, and a more summarized, quickly digestible option on the right. Perhaps both are available to the user, but the right side is what we are actively pushing learners to.

Review Lesson

Unify disparate sources of information into one intuitive lesson review UI, under lesson history



Receive Actionable Feedback

Cohesively synthesize all inputs into a consistent output, including feedback and recommendations

Feedback Report

Consistently formatted and “Camblyfied” tutor-generated, AI-assisted feedback,

Recommendations

Recommended snacktivities, quizzes, tutors, courses, and more based on feedback

To roll this product out (with GM support) we will need to:

- Adoption
- In-product education/tooltips
- Emails/notifs
- Targeting events

Anticipated Impact on Tutor Teams (Tutors to assess):

What is the impact on tutors’ day-to-day operations?

What is the impact on tutors’ accounting/financials?

What is the impact on tutor policy and/or guidance?

What tracking will the tutor team need? (Cambly admin view and Metabase)

What in product notifications would support tutors in this launch?

Anticipated Impact on Support Teams (Support to assess):

Using green/yellow/orange/red as signals, what is the anticipated impact on support?

- Impact on support load by ticket count, by ticket handle time, and by projected duration of impact. If the anticipated impact is yellow or higher, include on the experiment dashboard a query for tracking ticket volume by bucket, and share the dashboard with Product Ops.
- Are updates to the public Help Center, macros, or internal Confluence documentation required?
- Anticipated cost of make-goods, both free lessons and plan extensions. If over \$10k, Finance requests advance notice.
- Do we anticipate an impact to Cambly’s reputation, and do we accept that impact?
- Will support teams need new or modified tools to support customers with the change?

Potential Legal Questions or Concerns (PRIVILEGED)

Is there anything about this initiative that should be flagged for a more in-depth legal review? Some quick examples (not exhaustive!):

- Log-in flow to make sure users are properly agreeing to terms;
- Initiatives that give more/less control to tutors (e.g., pay, scheduling, curriculum)
- Use/sharing of user data (e.g., recordings of sessions, behavioral advertising, etc)
- Changes to our GDPR/CCPA/LGPD/etc data-deletion/export process?
- Copyright or trademark concerns

When in doubt, just ask Amber for a quick review.

Key Dates:

Launch date: 6/18/24

Target markets:

Global

Experiment size:

Experiment sequencing:

- Phase 1: launching the feature w/out driving the feature awareness / adoption, track the following:
 - [Implicit signals]
 - Frequency of students using the feature
 - Frequency of students watching the lesson video
 - [Explicit signals]
 - Could we ask students to thumb up / thumb down on this transcript feature?
 - Could we have them rate the accuracy / frequency of the fdbk?
- Phase 2: A/B test the feature by only driving feature awareness in the treatment group

Open Questions:

Result:

FAQ:

Notes:  **Notes - Corrective line edits weekly**

Appendix:

If we were to show *all* in line corrections, then something like the UI of a grammar checker

- Show in-line corrections
 - Something like the [UI of Grammarly would be nice](#), but it doesn't need to be quite so fancy. I like that the errors are underlined, for example, and the corrections and explanations are shown in context of the original text.

Demo document

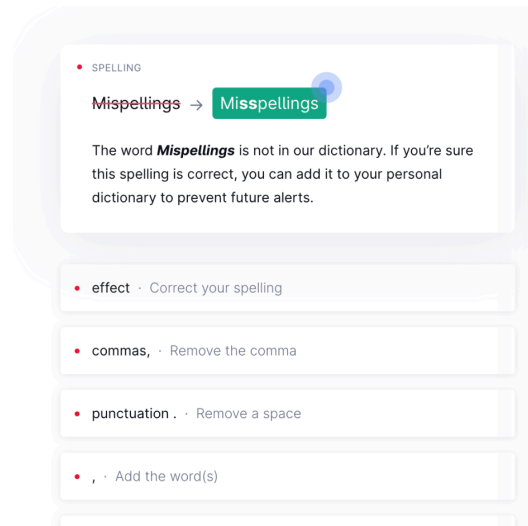
The basics

Misspellings and grammatical errors can effect your credibility. The same goes for misused commas, and other types of punctuation . Not only will Grammarly underline these issues in red, it will also showed you how to correctly write the sentence.

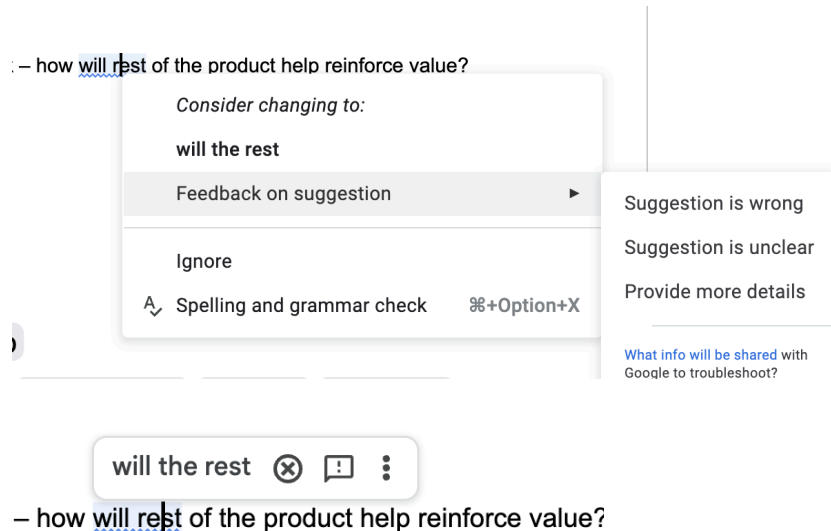
But wait...there's more?

Blue underlines mean a clarity issue has been spotted by Grammarly. You'll find suggestions that can possibly help you revise a wordy sentence in an effortless manner.

Grammarly will also inspect your vocabulary carefully and suggest the best word with **green** underlines to make sure you don't have to analyze your writing too much.



Google docs UI, for lighter lift inspo:



We spent a lot of time trying to figure out how to prompt the LLM to provide feedback assuming the following:

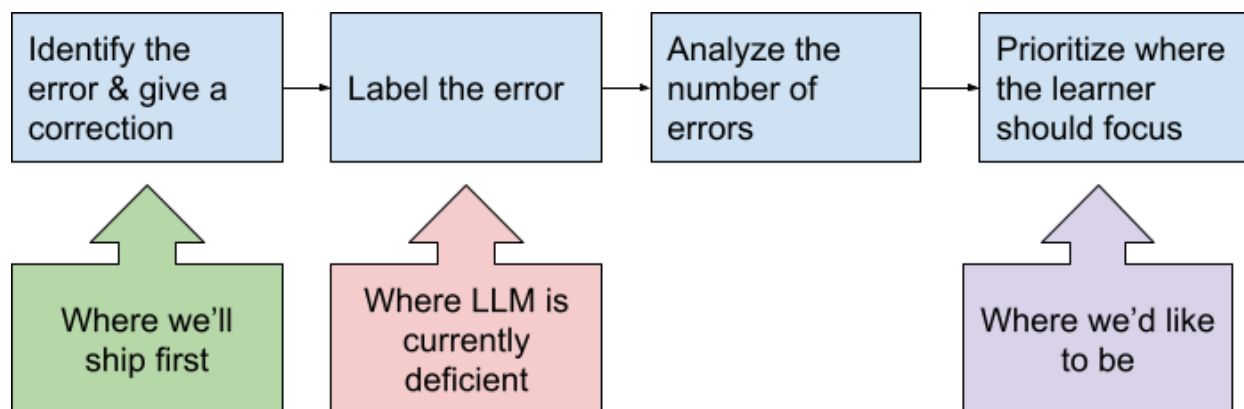
- It's more important to provide students with summarized, prioritized feedback that allows them to focus on the right areas
- Most tutors on Cambly will not be able to identify areas of grammar or vocabulary where the learner should focus.
 - As Cambly, we can supplement tutor feedback by correctly identifying grammar mistakes the learner makes consistently.

By exploring some different options, we have started circling around two distinct hypotheses:

Hypotheses (these are not mutually exclusive)	Product limitations	Mitigations / how we'll learn
<u>The on-the-ground view:</u> Students will find <i>all</i> corrective feedback helpful, even if we occasionally get it wrong	All feedback might be too randomizing, overwhelming	• Feature retention • Showing ≤ 10 corrections / explanations
	Line by line feedback will not be actionable for longer-term learning	This is a risk in the interim without a summarized solution. We will continue to build toward a solution that summarizes actionable feedback that will point learners toward the right direction for learning.
	Some feedback might be wrong, which could decrease trust or even be harmful because it misdirects learners.	• Feature retention • Rate the feedback • Guardrails around how often we get it wrong (and how do we measure?) • $< 50\%$ test population with a diverse set of goals, levels, and geos.
	Too many nitpicky corrections will be demotivating and detrimental to student learning	• Students need to opt-in to seeing corrections • We will experiment on giving students the option to see corrections to understand impact on downstream engagement
<u>The 30,000 ft view:</u> Students will find more benefit from summarized, prioritized feedback	We do not know yet how to prompt the LLM to provide accurate, summarized feedback that consolidates and prioritizes observations of what the learner should work on. This is primarily blocked on our ability to accurately label the error with the correct grammar / vocab rule.	Eng to continue to explore other tools and to prompt tune in partnership with Casey to identify a path forward

We aspire to provide a learner with precise, actionable feedback in order to continue with their learning. For example, rather than receiving feedback that they need to "work on their grammar," Cambly and/or the tutor provides feedback that the learner needs to work on "definite and indefinite articles."

We are building toward something like the below:



- **Identify the error & give a correction:** Learners are asking for help in identifying and correcting the errors. Let's build this, make it optional to consume, and see what it does for learning and retention.
 - This also is a basic building block for where we want to go in the future. Our ability to get to our desired end state requires us to correctly identify errors and provide corrections to the learner.
- **Label the error:** The team is currently exploring how to provide accurate labels for grammar and vocabulary rules where learners had errors. Our attempts with prompt-tuning the LLM still fall far short – from our spot testing the LLM cannot consistently, or accurately identify the grammar or vocabulary rule that is broken. *We need to figure out how to do this consistently and accurately.*
- **Analyze the number of errors:** This should be a trivial amount of effort – the most basic version of this might just be a count of errors and making some recommendations about where the learner should focus given the number and frequency of errors.
- **Prioritize where the learner should focus:** The natural offshoot of the analysis in the prior step will be a precise recommendation around where the learner should focus, along with recommendations of activities they can do (lessons, snacks, etc.)